Nifty Futures — Extended Backtest 2013–2026: Popular Strategies, Trend(50) + Rising OI, Re-tuning and Regimes

Research only, not investment advice. Prepared 2026-11-03. **All results are on real NSE Nifty index futures** (daily bhavcopy). Spot is not used. This report extends the earlier 2018–2026 report with **1,292 additional trading days (2012-01-02 to 2017-03-31)**, so the data now covers 2012–2026 (3659 trading days; 178 expiries).

1. Verdict (short)

Hinglish mein: 2013–2017 ka naya data aane ke baad bhi Trend(50)+rising OI strategy buy-and-hold se behtar nahi nikli. 2013–17 mein (jo data pehle kabhi nahi dekha tha) uska result -2.2% CAGR raha, jabki buy-and-hold 4.5% tha. Pichle test period mein bhi -8.5% tha. Tune karne par bhi koi trend/OI strategy test mein paisa nahi bana payi. Ek hi family jo har period mein positive rahi: IBS mean-reversion (3 se 6% CAGR, drawdown -6% se -14%, buy-and-hold ka -41%). Ye chhota edge hai, aur trials ke hisaab se statistically proven nahi hai (Deflated Sharpe 0.10).

In English:

- **Trend(50) + rising OI still fails.** On the never-seen 2013–17 window it returned **-2.2% CAGR / -0.15 Sharpe / -27.0% MaxDD** against buy-and-hold's **+4.5% / 0.33 / -27.4%**. Combined with the earlier test-period loss (-8.5% vs -4.8%), it has now failed on two independent windows.
- **Fine-tuning did not rescue it.** Re-tuning on the longer 2013–21 train and checking all five parameters (N=30/40/50/60/80) gives no N that beats buy-and-hold in both the 2013–17 and the 2024–26 windows.
- **No trend-following, breakout or OI strategy was profitable in all four windows** (2013–17, 2013–21 train, 2022–23, 2024–26). Trend(50)+rising OI earned money in 2020–2023 and lost in 2013 (-11.2%), 2016 (-6.6%) and 2024 (-14.5%).
- **One family stayed positive everywhere: IBS mean-reversion** (long when the close is in the bottom of the day's range). Parameters frozen from 2018–21 earn **6.0% in 2013–17, 3.6% (2013–21), 1.1% (2022–23) and 3.1% (2024–26)**, with drawdowns of -5.6% to -13.8%, against buy-and-hold's -41%. It is a small edge. It is positive in 10 of 14 years, but it does not beat buy-and-hold in strong bull years (2014, 2017, 2020, 2021, 2023).
- **A VIX filter on IBS (hold only when India VIX is at least 15) lifts it to 7.4% / 2.8% / 2.1% / 5.5% in the four windows (full 2013–26: 4.3% CAGR, Sharpe 0.65, MaxDD -12.2% vs buy-and-hold 4.6% / 0.35 / -40.7%).** I chose the filter after seeing the regime tables, so it is partly in-sample (section 10.5).
- **It is not statistically proven.** After counting about 2,700 trials, its Deflated Sharpe probability is 0.10 (0.95+ would be strong evidence), and nearby parameters (lo=0.2 and above) fail in the test period.
- **Walk-forward re-tuning every year (2015–2026) earns 0.6% CAGR** against buy-and-hold's 3.6%, so automatic yearly re-selection from this strategy menu does not work.

2. Data and method

Item	Choice
New data	NSE F&O daily bhavcopy for 2012-01-02 to 2017-03-31, fetched through your own Chrome session (NSE's archive blocks scripted clients). 1,292 trading days, 3 Nifty futures contracts per day, 78 non-trading dates returned 404 (holidays). India VIX for 2012-01-02 to 2016-12-30 from NSE's historical VIX page (1,240 rows). Quality checks: no zero prices, no duplicates, largest one-day front-month move 6.0% (24 Aug 2015).
Instrument	Nifty index futures, front-month contract not expiring on the entry day; rolled the day it expires
Signal timing	Computed from data through the close of day t, including that day's price, VIX and OI. Orders fill at the **next open**; held open-to-open on the same contract.

Item	Choice
Costs	Brokerage, STT (changes by date, 0.017% to 0.02%), exchange, SEBI, GST, stamp, 0.01% slippage per side, plus a roll cost. Section 6.
Carry / interest	Futures prices already include the premium, so longs pay it and the buy-and-hold benchmark pays it too. No interest is added. About 6.5%/yr on idle capital would be added to every series equally.
Warm-up	First year (2012) is indicator warm-up. Evaluation starts 2013-01-01.
Windows	**Pre 2013-01 to 2017-03** (new), **Train 2013-01 to 2021-12**, **Valid 2022-01 to 2023-12**, **Test 2024-01 to 2026-11**, **Full 2013-2026**. The old train (2018-04 to 2021-12) is also kept for the frozen-parameter check.
Two-stage protocol	**Stage 1:** parameters frozen from the earlier report (tuned on 2018–21 only) run unchanged on 2013–17, which they had never seen. **Stage 2:** re-tune on the extended train 2013–21, shortlist on train+valid, show test once.
Benchmark	Always-long Nifty futures with the same costs and roll rule

3. Stage 1 — frozen parameters on the never-seen 2013–17 window (CAGR% / Sharpe / MaxDD%)

These parameters were chosen on 2018–21 only. 2013–17 was not touched, so this is a clean out-of-sample test.

Strategy (params frozen from 2018-21 train)	Pre 2013-17 (never seen)	Old train 2018-21	Valid 2022-23	Test 2024-26	Beats B&H on pre?
IBS mean-reversion {'lo': 0.1, 'hi': 0.8}	**6.0 / 0.82 / -5.6**	2.6 / 0.31 / -11.7	1.1 / 0.25 / -6.1	3.1 / 0.50 / -7.9	YES
Stochastic mean-rev (14,20/80) {'w': 14, 'lo': 30, 'hi': 70}	**6.9 / 0.61 / -15.4**	3.7 / 0.31 / -33.8	-1.9 / -0.13 / -11.6	1.1 / 0.16 / -13.4	YES
Buy-the-dip (-1.5 sigma) {'th': 1.0}	**5.1 / 0.58 / -15.0**	1.8 / 0.25 / -11.2	-0.5 / -0.04 / -6.9	3.2 / 0.47 / -8.5	YES
RSI(14) 30/50 mean-rev {'w': 21, 'lo': 30, 'hi': 60}	**1.5 / 0.56 / -4.3**	0.5 / 0.11 / -31.5	0.0 / 0.00 / 0.0	-4.4 / -0.42 / -18.8	no
EMA cross 12/26 [long-only] {'a': 12, 'b': 26}	**5.7 / 0.47 / -21.2**	14.9 / 1.23 / -13.1	6.9 / 0.79 / -7.9	-5.2 / -0.51 / -17.3	YES
Supertrend daily (10,3) [long-only] {'w': 7, 'm': 2.0}	**5.2 / 0.44 / -22.5**	12.4 / 1.13 / -13.2	13.8 / 1.43 / -11.1	-6.1 / -0.71 / -19.1	YES
Long build-up (price&OI up over k) [expiry-adj OI] {'k': 20}	**4.8 / 0.41 / -20.1**	11.2 / 0.97 / -13.1	8.5 / 0.98 / -10.9	-9.4 / -1.12 / -25.7	YES
EMA cross 12/26 [L/S] {'a': 12, 'b': 26}	**4.7 / 0.34 / -30.7**	15.3 / 0.81 / -22.9	5.7 / 0.46 / -10.0	-7.2 / -0.46 / -25.2	no
ADX/DI trend (14,25) [long-only] {'w': 14, 'th': 30}	**1.3 / 0.32 / -6.0**	3.1 / 0.61 / -5.9	1.7 / 0.43 / -4.3	-0.5 / -0.27 / -2.8	no
Supertrend daily (10,3) [L/S] {'w': 7, 'm': 2.0}	**3.7 / 0.29 / -32.9**	10.1 / 0.58 / -23.1	19.8 / 1.33 / -11.5	-9.3 / -0.62 / -30.2	no
Long build-up (price&OI up over k) [naive OI] {'k': 20}	**1.4 / 0.21 / -10.6**	4.9 / 0.63 / -9.1	5.6 / 0.93 / -4.4	-4.9 / -0.78 / -18.5	no
Connors RSI(2) 10/70 {'w': 4, 'lo': 10, 'hi': 80}	**0.9 / 0.19 / -7.8**	5.2 / 1.46 / -0.4	1.0 / 0.18 / -11.6	-3.0 / -0.61 / -12.7	no
MACD 12/26/9 [long-only] {'a': 16, 'b': 40, 's': 9}	**0.5 / 0.11 / -23.2**	15.9 / 1.30 / -10.3	12.6 / 1.37 / -9.8	-9.1 / -1.05 / -25.2	no
Price > 200DMA [long-only] {'w': 50}	**0.5 / 0.11 / -23.2**	15.1 / 1.31 / -12.6	1.6 / 0.22 / -16.8	-4.6 / -0.50 / -15.7	no
Trend(50) only [no OI] {'w': 50}	**0.5 / 0.11 / -23.2**	15.1 / 1.31 / -12.6	1.6 / 0.22 / -16.8	-4.6 / -0.50 / -15.7	no
Keltner breakout (20,2) {'w': 20, 'm': 1.5}	**0.4 / 0.09 / -13.8**	8.8 / 0.98 / -12.7	6.7 / 0.95 / -7.2	-7.0 / -1.20 / -19.1	no
Trend(N)+OI sma20 [expiry-adj OI] {'N': 50}	**0.4 / 0.09 / -23.2**	13.5 / 1.22 / -13.3	5.8 / 0.72 / -10.9	-3.3 / -0.36 / -11.6	no
Trend(N)+OI sma20 [naive OI] {'N': 50}	**0.4 / 0.09 / -12.7**	4.3 / 0.55 / -9.0	1.3 / 0.23 / -7.8	-0.3 / -0.02 / -6.0	no
SMA cross 20/50 [long-only] {'a': 30, 'b': 50}	**0.1 / 0.06 / -22.7**	17.5 / 1.35 / -12.2	-7.8 / -0.79 / -18.3	-0.8 / -0.04 / -11.4	no
Bollinger breakout (20,2) {'w': 30, 'k': 2.5}	**-0.1 / 0.00 / -11.9**	9.1 / 1.14 / -9.1	1.9 / 0.69 / -2.3	-1.3 / -0.69 / -4.7	no
SMA cross 50/200 [long-only] {'a': 20, 'b': 100}	**-1.3 / -0.07 / -22.6**	11.2 / 0.94 / -15.8	0.3 / 0.08 / -11.4	0.6 / 0.11 / -10.8	no
Trend(N)+OI chg5 [naive OI] {'N': 50}	**-1.4 / -0.11 / -17.6**	4.2 / 0.51 / -9.9	5.5 / 0.93 / -3.6	-3.5 / -0.57 / -12.0	no
Time-series momentum 252d [long-only] {'w': 63}	**-2.0 / -0.13 / -23.8**	10.3 / 0.89 / -12.8	5.7 / 0.66 / -9.3	-6.0 / -0.69 / -18.3	no
Trend(N)+OI chg5 [expiry-adj OI] {'N': 50}	**-2.2 / -0.15 / -27.0**	11.3 / 1.11 / -11.7	10.1 / 1.47 / -4.4	-8.5 / -1.15 / -23.9	no
MACD 12/26/9 [L/S] {'a': 16, 'b': 40, 's': 9}	**-5.5 / -0.21 / -37.2**	17.2 / 0.89 / -14.6	17.4 / 1.19 / -11.0	-14.9 / -1.08 / -41.0	no

Strategy (params frozen from 2018-21 train)	Pre 2013-17 (never seen)	Old train 2018-21	Valid 2022-23	Test 2024-26	Beats B&H on pre?
Price > 200DMA [L/S] {'w': 50}	**-5.6 / -0.22 / -43.5**	15.6 / 0.82 / -22.2	-4.4 / -0.24 / -23.0	-6.2 / -0.39 / -23.5	no
Trend(N)+OI chg1 [naive OI] {'N': 50}	**-2.4 / -0.22 / -18.9**	8.1 / 0.90 / -11.0	-1.9 / -0.23 / -11.2	-0.8 / -0.10 / -10.9	no
Bollinger mean-rev (20,2) {'w': 10, 'k': 2.5}	**-0.6 / -0.24 / -6.3**	0.2 / 0.70 / -0.1	0.0 / 0.00 / 0.0	0.0 / 0.00 / 0.0	no
SMA cross 20/50 [L/S] {'a': 30, 'b': 50}	**-6.7 / -0.28 / -44.8**	20.9 / 1.05 / -22.2	-21.2 / -1.59 / -39.1	1.6 / 0.18 / -13.5	no
Donchian 20/10 (turtle) {'en': 40, 'ex': 10}	**-3.0 / -0.36 / -24.1**	8.6 / 0.91 / -13.7	0.1 / 0.05 / -12.3	-3.8 / -0.53 / -13.7	no
Donchian 55/20 {'en': 40, 'ex': 10}	**-3.0 / -0.36 / -24.1**	8.6 / 0.91 / -13.7	0.1 / 0.05 / -12.3	-3.8 / -0.53 / -13.7	no
ADX/DI trend (14,25) [L/S] {'w': 14, 'th': 30}	**-3.5 / -0.38 / -18.6**	5.5 / 0.43 / -19.0	-4.7 / -0.77 / -12.4	-2.3 / -0.25 / -10.2	no
Trend(N)+OI chg1 [expiry-adj OI] {'N': 50}	**-4.5 / -0.38 / -26.2**	9.7 / 1.01 / -10.8	0.4 / 0.10 / -10.7	-2.9 / -0.42 / -12.3	no
SMA cross 50/200 [L/S] {'a': 20, 'b': 100}	**-9.4 / -0.44 / -44.2**	8.0 / 0.48 / -29.0	-6.8 / -0.42 / -21.7	4.3 / 0.37 / -15.8	no
Time-series momentum 252d [L/S] {'w': 63}	**-10.7 / -0.52 / -47.6**	6.3 / 0.40 / -27.3	3.4 / 0.31 / -18.8	-9.0 / -0.60 / -26.0	no
Buy & Hold futures	**4.5 / 0.33 / -27.4**	11.0 / 0.62 / -40.7	6.0 / 0.48 / -17.8	-4.8 / -0.28 / -25.9	no

Reading it: the trend and OI strategies that looked best in 2018–23 mostly **did not beat buy-and-hold in 2013–17** (Trend(50)+rising OI: -2.2% vs 4.5%). The strategies that did beat it there were mean-reversion ones (IBS, Stochastic, buy-the-dip) and some trend variants that then lost in 2024–26 (EMA, Supertrend, long build-up).

4. Stage 2 — re-tuned on the extended train (2013–2021)

4.1 Textbook parameters, no fitting

Strategy	Pre 2013-17	Train 2013-21	Valid 2022-23	Test 2024-26	Full 2013-26	Exposure
EMA cross 12/26 [long-only]	5.7 / 0.47 / -21.2	9.0 / 0.75 / -21.2	6.9 / 0.79 / -7.9	-5.2 / -0.51 / -17.3	5.6 / 0.53 / -21.2	60%
Connors RSI(2) 10/70	3.5 / 0.60 / -6.2	3.1 / 0.48 / -13.3	-1.8 / -0.31 / -7.0	4.6 / 0.97 / -3.1	2.7 / 0.45 / -13.3	15%
Trend(N)+OI sma20 [expiry-adj OI]	0.4 / 0.09 / -23.2	6.3 / 0.60 / -23.2	5.8 / 0.72 / -10.9	-3.3 / -0.36 / -11.6	4.2 / 0.45 / -23.2	55%
Trend(50) only [no OI]	0.5 / 0.11 / -23.2	7.1 / 0.66 / -23.2	1.6 / 0.22 / -16.8	-4.6 / -0.50 / -15.7	3.8 / 0.40 / -23.2	59%
IBS mean-reversion	13.7 / 1.08 / -10.2	6.8 / 0.55 / -32.9	-3.1 / -0.28 / -9.4	-0.9 / -0.04 / -13.1	3.7 / 0.35 / -32.9	40%
Buy & Hold futures	4.5 / 0.33 / -27.4	7.4 / 0.48 / -40.7	6.0 / 0.48 / -17.8	-4.8 / -0.28 / -25.9	4.6 / 0.35 / -40.7	100%
Bollinger mean-rev (20,2)	3.1 / 0.38 / -8.8	2.9 / 0.31 / -31.5	2.4 / 0.33 / -5.8	4.5 / 0.54 / -10.3	3.1 / 0.35 / -31.5	22%
EMA cross 12/26 [L/S]	4.7 / 0.34 / -30.7	8.1 / 0.51 / -30.7	5.7 / 0.46 / -10.0	-7.2 / -0.46 / -25.2	4.4 / 0.34 / -30.7	100%
SMA cross 20/50 [long-only]	-1.0 / -0.04 / -24.7	5.9 / 0.56 / -24.7	-4.4 / -0.44 / -18.0	-0.6 / -0.02 / -10.8	3.0 / 0.33 / -24.7	61%
Supertrend daily (10,3) [long-only]	1.9 / 0.21 / -21.9	6.3 / 0.56 / -21.9	10.2 / 1.20 / -7.0	-12.5 / -1.36 / -32.0	2.6 / 0.29 / -32.0	57%
Bollinger breakout (20,2)	2.7 / 0.30 / -17.2	4.3 / 0.50 / -17.2	8.8 / 1.43 / -4.3	-10.0 / -2.04 / -25.5	1.8 / 0.26 / -25.8	24%
Price > 200DMA [long-only]	1.9 / 0.20 / -25.9	5.4 / 0.48 / -25.9	-1.9 / -0.14 / -17.4	-4.2 / -0.43 / -22.1	2.3 / 0.25 / -25.9	64%
Trend(N)+OI sma20 [naive OI]	0.4 / 0.09 / -12.7	2.2 / 0.31 / -12.7	1.3 / 0.23 / -7.8	-0.3 / -0.02 / -6.0	1.6 / 0.25 / -12.7	30%
Trend(N)+OI chg5 [expiry-adj OI]	-2.2 / -0.15 / -27.0	3.6 / 0.40 / -27.0	10.1 / 1.47 / -4.4	-8.5 / -1.15 / -23.9	1.9 / 0.25 / -27.0	45%
Keltner breakout (20,2)	1.3 / 0.21 / -11.7	3.7 / 0.51 / -11.7	3.5 / 0.56 / -9.6	-6.6 / -1.20 / -18.3	1.4 / 0.24 / -18.3	29%
MACD 12/26/9 [long-only]	-0.7 / 0.01 / -21.8	2.8 / 0.28 / -21.8	9.9 / 1.10 / -10.9	-5.8 / -0.63 / -17.4	1.9 / 0.23 / -21.8	48%
Trend(N)+OI chg1 [naive OI]	-2.4 / -0.22 / -18.9	3.0 / 0.38 / -18.9	-1.9 / -0.23 / -11.2	-0.8 / -0.10 / -10.9	1.5 / 0.22 / -18.9	36%
Stochastic mean-rev (14,20/80)	5.4 / 0.48 / -13.6	3.6 / 0.32 / -33.9	-1.7 / -0.11 / -11.9	-0.8 / -0.03 / -16.6	1.9 / 0.21 / -33.9	42%

Strategy	Pre 2013-17	Train 2013-21	Valid 2022-23	Test 2024-26	Full 2013-26	Exposure
SMA cross 50/200 [long-only]	2.8 / 0.27 / -19.0	3.0 / 0.28 / -39.6	0.8 / 0.12 / -14.9	-0.9 / -0.04 / -18.1	1.9 / 0.21 / -39.6	64%
RSI(14) 30/50 mean-rev	3.6 / 0.44 / -8.8	0.7 / 0.12 / -31.7	6.2 / 1.23 / -4.5	-0.9 / -0.07 / -11.0	1.1 / 0.17 / -31.7	14%
Trend(N)+OI chg1 [expiry-adj OI]	-4.5 / -0.38 / -26.2	2.6 / 0.31 / -26.2	0.4 / 0.10 / -10.7	-2.9 / -0.42 / -12.3	1.1 / 0.17 / -26.2	39%
Long build-up (price&OI up over k) [expiry-adj OI]	-0.2 / 0.05 / -20.7	2.7 / 0.29 / -20.7	12.0 / 1.56 / -7.8	-10.1 / -1.54 / -26.5	1.2 / 0.17 / -27.3	44%
Donchian 20/10 (turtle)	-0.6 / 0.01 / -22.3	3.0 / 0.32 / -22.3	6.4 / 0.82 / -7.9	-8.4 / -1.06 / -22.6	1.0 / 0.15 / -22.6	43%
Trend(N)+OI chg5 [naive OI]	-1.4 / -0.11 / -17.6	1.1 / 0.17 / -17.6	5.5 / 0.93 / -3.6	-3.5 / -0.57 / -12.0	0.7 / 0.13 / -17.6	34%
Buy-the-dip (-1.5 sigma)	-0.2 / -0.02 / -12.0	-0.4 / -0.04 / -20.6	-1.3 / -0.25 / -6.8	4.1 / 0.73 / -5.5	0.4 / 0.11 / -20.6	7%
SMA cross 20/50 [L/S]	-8.8 / -0.41 / -48.4	1.8 / 0.19 / -48.4	-15.5 / -1.10 / -32.8	2.0 / 0.21 / -16.9	-0.8 / 0.04 / -48.4	100%
Donchian 55/20	-0.2 / 0.02 / -14.9	1.1 / 0.17 / -18.3	2.5 / 0.42 / -8.8	-5.4 / -0.72 / -15.8	-0.1 / 0.04 / -18.3	41%
ADX/DI trend (14,25) [long-only]	-1.1 / -0.17 / -17.6	0.6 / 0.13 / -17.6	0.1 / 0.04 / -5.8	-1.6 / -0.55 / -6.6	0.1 / 0.04 / -17.6	19%
Time-series momentum 252d [long-only]	-0.6 / 0.01 / -23.1	2.9 / 0.30 / -23.1	-2.8 / -0.16 / -18.8	-8.2 / -0.76 / -31.6	-0.3 / 0.03 / -31.8	68%
Supertrend daily (10,3) [L/S]	-2.6 / -0.05 / -34.2	2.6 / 0.23 / -34.2	12.1 / 0.87 / -12.4	-21.1 / -1.62 / -51.9	-1.5 / -0.00 / -51.9	100%
Price > 200DMA [L/S]	-2.6 / -0.05 / -37.6	1.2 / 0.16 / -37.6	-10.6 / -0.71 / -31.1	-5.3 / -0.32 / -34.3	-1.9 / -0.03 / -44.4	100%
SMA cross 50/200 [L/S]	-0.8 / 0.05 / -25.8	-3.1 / -0.08 / -59.7	-5.5 / -0.32 / -28.6	1.5 / 0.17 / -23.0	-2.5 / -0.06 / -59.7	100%
MACD 12/26/9 [L/S]	-7.8 / -0.35 / -43.0	-4.3 / -0.14 / -46.9	11.7 / 0.84 / -15.9	-8.5 / -0.56 / -33.3	-3.0 / -0.09 / -53.2	100%
Long build-up (price&OI up over k) [naive OI]	-2.7 / -0.25 / -15.2	-0.7 / -0.03 / -15.2	5.8 / 0.93 / -9.4	-7.1 / -1.48 / -18.6	-1.1 / -0.11 / -20.3	32%
ADX/DI trend (14,25) [L/S]	-7.9 / -0.71 / -37.2	-3.0 / -0.17 / -37.2	-5.6 / -0.61 / -17.7	-0.3 / 0.02 / -10.9	-2.8 / -0.19 / -37.2	38%
Time-series momentum 252d [L/S]	-7.5 / -0.33 / -38.9	-3.5 / -0.10 / -51.1	-11.7 / -0.79 / -30.7	-12.8 / -0.91 / -45.0	-6.7 / -0.32 / -66.8	100%

4.2 Parameters re-tuned on train 2013–21 only

Strategy (params tuned on TRAIN 2013-21)	Pre 2013-17	Train 2013-21	Valid 2022-23	Test 2024-26	Full 2013-26	Exposure
Bollinger breakout (20,2) {'w': 50, 'k': 2.0}	5.1 / 0.66 / -10.7	6.6 / 0.79 / -10.7	1.6 / 0.32 / -5.6	-1.1 / -0.13 / -9.5	4.2 / 0.57 / -10.7	35%
EMA cross 12/26 [long-only] {'a': 20, 'b': 21}	5.3 / 0.52 / -11.6	8.2 / 0.76 / -15.1	4.6 / 0.55 / -8.2	-4.6 / -0.44 / -17.5	4.9 / 0.51 / -17.5	60%
Supertrend daily (10,3) [long-only] {'w': 7, 'm': 2.0}	5.2 / 0.44 / -22.5	8.1 / 0.71 / -22.5	13.8 / 1.43 / -11.1	-6.1 / -0.71 / -19.1	5.8 / 0.57 / -22.5	55%
Connors RSI(2) 10/70 {'w': 4, 'lo': 10, 'hi': 80}	0.9 / 0.19 / -7.8	3.4 / 0.71 / -7.8	1.0 / 0.18 / -11.6	-3.0 / -0.61 / -12.7	1.7 / 0.35 / -13.5	10%
SMA cross 20/50 [long-only] {'a': 15, 'b': 80}	1.4 / 0.18 / -18.8	7.6 / 0.71 / -18.8	2.0 / 0.26 / -10.7	-0.8 / -0.05 / -12.7	5.0 / 0.52 / -18.8	61%
Price > 200DMA [long-only] {'w': 50}	0.5 / 0.11 / -23.2	7.1 / 0.66 / -23.2	1.6 / 0.22 / -16.8	-4.6 / -0.50 / -15.7	3.8 / 0.40 / -23.2	59%
Trend(50) only [no OI] {'w': 50}	0.5 / 0.11 / -23.2	7.1 / 0.66 / -23.2	1.6 / 0.22 / -16.8	-4.6 / -0.50 / -15.7	3.8 / 0.40 / -23.2	59%
Long build-up (price&OI up over k) [expiry-adj OI] {'k': 20}	4.8 / 0.41 / -20.1	7.0 / 0.61 / -20.1	8.5 / 0.98 / -10.9	-9.4 / -1.12 / -25.7	3.6 / 0.37 / -25.7	55%
Trend(N)+OI sma20 [expiry-adj OI] {'N': 50}	0.4 / 0.09 / -23.2	6.3 / 0.60 / -23.2	5.8 / 0.72 / -10.9	-3.3 / -0.36 / -11.6	4.2 / 0.45 / -23.2	55%
MACD 12/26/9 [long-only] {'a': 16, 'b': 40, 's': 9}	0.5 / 0.11 / -23.2	6.6 / 0.59 / -23.2	12.6 / 1.37 / -9.8	-9.1 / -1.05 / -25.2	4.0 / 0.41 / -25.5	49%
IBS mean-reversion {'lo': 0.1, 'hi': 0.7}	7.5 / 1.14 / -5.8	4.1 / 0.56 / -16.9	0.1 / 0.04 / -6.1	1.1 / 0.21 / -8.1	2.9 / 0.44 / -16.9	13%
EMA cross 12/26 [L/S] {'a': 12, 'b': 26}	4.7 / 0.34 / -30.7	8.1 / 0.51 / -30.7	5.7 / 0.46 / -10.0	-7.2 / -0.46 / -25.2	4.4 / 0.34 / -30.7	100%
Keltner breakout (20,2) {'w': 20, 'm': 2.0}	1.3 / 0.21 / -11.7	3.7 / 0.51 / -11.7	3.5 / 0.56 / -9.6	-6.6 / -1.20 / -18.3	1.4 / 0.24 / -18.3	29%
Stochastic mean-rev (14,20/80) {'w': 14, 'lo': 30, 'hi': 70}	6.9 / 0.61 / -15.4	6.0 / 0.50 / -33.8	-1.9 / -0.13 / -11.6	1.1 / 0.16 / -13.4	3.8 / 0.36 / -33.8	43%
ADX/DI trend (14,25) [long-only] {'w': 20, 'th': 20}	2.3 / 0.40 / -9.6	2.7 / 0.46 / -10.4	-0.6 / -0.08 / -6.2	-0.6 / -0.18 / -4.9	1.5 / 0.30 / -14.2	20%
SMA cross 50/200 [long-only] {'a': 20, 'b': 100}	-1.3 / -0.07 / -22.6	4.7 / 0.46 / -22.6	0.3 / 0.08 / -11.4	0.6 / 0.11 / -10.8	3.2 / 0.35 / -22.6	62%

Strategy (params tuned on TRAIN 2013-21)	Pre 2013-17	Train 2013-21	Valid 2022-23	Test 2024-26	Full 2013-26	Exposure
Trend(N)+OI chg5 [expiry-adj OI] {'N': 40}	2.2 / 0.23 / -25.9	4.3 / 0.44 / -25.9	10.8 / 1.55 / -4.5	-9.5 / -1.22 / -26.6	2.2 / 0.27 / -26.6	45%
Supertrend daily (10,3) [L/S] {'w': 7, 'm': 2.0}	3.7 / 0.29 / -32.9	6.1 / 0.41 / -32.9	19.8 / 1.33 / -11.5	-9.3 / -0.62 / -30.2	4.6 / 0.35 / -32.9	100%
Donchian 55/20 {'en': 80, 'ex': 40}	3.0 / 0.36 / -9.1	3.6 / 0.41 / -22.4	-1.0 / -0.11 / -9.3	-1.5 / -0.16 / -8.3	1.8 / 0.25 / -22.4	46%
Time-series momentum 252d [long-only] {'w': 63}	-2.0 / -0.13 / -23.8	3.8 / 0.39 / -23.8	5.7 / 0.66 / -9.3	-6.0 / -0.69 / -18.3	2.0 / 0.24 / -23.8	62%
SMA cross 20/50 [L/S] {'a': 30, 'b': 50}	-6.7 / -0.28 / -44.8	5.6 / 0.39 / -44.8	-21.2 / -1.59 / -39.1	1.6 / 0.18 / -13.5	0.4 / 0.11 / -48.5	100%
Trend(N)+OI chg1 [naive OI] {'N': 50}	-2.4 / -0.22 / -18.9	3.0 / 0.38 / -18.9	-1.9 / -0.23 / -11.2	-0.8 / -0.10 / -10.9	1.5 / 0.22 / -18.9	36%
Donchian 20/10 (turtle) {'en': 20, 'ex': 20}	3.3 / 0.30 / -21.7	3.7 / 0.35 / -23.1	2.8 / 0.36 / -11.2	-9.7 / -1.17 / -26.2	0.7 / 0.12 / -26.7	54%
Buy-the-dip (-1.5 sigma) {'th': 1.0}	5.1 / 0.58 / -15.0	2.6 / 0.33 / -19.7	-0.5 / -0.04 / -6.9	3.2 / 0.47 / -8.5	2.3 / 0.31 / -19.7	16%
Trend(N)+OI sma20 [naive OI] {'N': 50}	0.4 / 0.09 / -12.7	2.2 / 0.31 / -12.7	1.3 / 0.23 / -7.8	-0.3 / -0.02 / -6.0	1.6 / 0.25 / -12.7	30%
Trend(N)+OI chg1 [expiry-adj OI] {'N': 50}	-4.5 / -0.38 / -26.2	2.6 / 0.31 / -26.2	0.4 / 0.10 / -10.7	-2.9 / -0.42 / -12.3	1.1 / 0.17 / -26.2	39%
Bollinger mean-rev (20,2) {'w': 20, 'k': 2.0}	3.1 / 0.38 / -8.8	2.9 / 0.31 / -31.5	2.4 / 0.33 / -5.8	4.5 / 0.54 / -10.3	3.1 / 0.35 / -31.5	22%
Price > 200DMA [L/S] {'w': 50}	-5.6 / -0.22 / -43.5	4.0 / 0.30 / -43.5	-4.4 / -0.24 / -23.0	-6.2 / -0.39 / -23.5	0.6 / 0.12 / -43.5	100%
Long build-up (price&OI up over k) [naive OI] {'k': 20}	1.4 / 0.21 / -10.6	2.0 / 0.29 / -10.6	5.6 / 0.93 / -4.4	-4.9 / -0.78 / -18.5	1.1 / 0.18 / -18.5	30%
MACD 12/26/9 [L/S] {'a': 16, 'b': 40, 's': 9}	-5.5 / -0.21 / -37.2	3.0 / 0.25 / -40.4	17.4 / 1.19 / -11.0	-14.9 / -1.08 / -41.0	0.9 / 0.14 / -41.0	100%
RSI(14) 30/50 mean-rev {'w': 7, 'lo': 35, 'hi': 50}	3.1 / 0.36 / -10.6	1.7 / 0.20 / -35.5	-3.4 / -0.31 / -13.6	1.7 / 0.23 / -12.6	0.9 / 0.14 / -35.5	29%
Trend(N)+OI chg5 [naive OI] {'N': 40}	0.1 / 0.06 / -16.2	1.2 / 0.18 / -16.2	5.9 / 0.99 / -4.6	-4.4 / -0.73 / -14.6	0.7 / 0.12 / -16.2	34%
ADX/DI trend (14,25) [L/S] {'w': 14, 'th': 30}	-3.5 / -0.38 / -18.6	0.3 / 0.09 / -19.0	-4.7 / -0.77 / -12.4	-2.3 / -0.25 / -10.2	-1.0 / -0.04 / -28.4	23%
SMA cross 50/200 [L/S] {'a': 20, 'b': 100}	-9.4 / -0.44 / -44.2	-0.6 / 0.06 / -44.2	-6.8 / -0.42 / -21.7	4.3 / 0.37 / -15.8	-0.5 / 0.06 / -44.2	100%
Time-series momentum 252d [L/S] {'w': 63}	-10.7 / -0.52 / -47.6	-2.2 / -0.03 / -47.6	3.4 / 0.31 / -18.8	-9.0 / -0.60 / -26.0	-2.9 / -0.08 / -47.6	100%

Scoreboard: does the tuned strategy beat buy-and-hold on CAGR+Sharpe+MaxDD, and is its CAGR positive, in Pre/Train/Valid/Test?

Strategy (tuned 2013-21)	Beats B&H (CAGR+Sharpe+MaxDD) Pre/Train/Valid/Test	Positive CAGR Pre/Train/Valid/Test
ADX/DI trend (14,25) [L/S]	n/n/n/Y	-/+/-/-
ADX/DI trend (14,25) [long-only]	n/n/n/Y	+/+/-/-
Bollinger breakout (20,2)	Y/n/n/Y	+/+/+/-
Bollinger mean-rev (20,2)	n/n/n/Y	+/+/+/+
Buy-the-dip (-1.5 sigma)	Y/n/n/Y	+/+/-/+
Connors RSI(2) 10/70	n/n/n/n	+/+/+/-
Donchian 20/10 (turtle)	n/n/n/n	+/+/+/-
Donchian 55/20	n/n/n/Y	+/+/-/-
EMA cross 12/26 [L/S]	n/Y/n/n	+/+/+/-
EMA cross 12/26 [long-only]	Y/Y/n/n	+/+/+/-
IBS mean-reversion	Y/n/n/Y	+/+/+/+
Keltner breakout (20,2)	n/n/n/n	+/+/+/-
Long build-up (price&OI up over k) [expiry-adj OI]	Y/n/Y/n	+/+/+/-
Long build-up (price&OI up over k) [naive OI]	n/n/n/n	+/+/+/-
MACD 12/26/9 [L/S]	n/n/Y/n	-/+/+/-

Strategy (tuned 2013-21)	Beats B&H (CAGR+Sharpe+MaxDD) Pre/Train/Valid/Test	Positive CAGR Pre/Train/Valid/Test
MACD 12/26/9 [long-only]	n/n/Y/n	+/+/+/-
Price > 200DMA [L/S]	n/n/n/n	-/+/-/-
Price > 200DMA [long-only]	n/n/n/n	+/+/+/-
RSI(14) 30/50 mean-rev	n/n/n/Y	+/+/-/+
SMA cross 20/50 [L/S]	n/n/n/Y	-/+/-/+
SMA cross 20/50 [long-only]	n/Y/n/Y	+/+/+/-
SMA cross 50/200 [L/S]	n/n/n/Y	-/-/-/+
SMA cross 50/200 [long-only]	n/n/n/Y	-/+/+/+
Stochastic mean-rev (14,20/80)	Y/n/n/Y	+/+/-/+
Supertrend daily (10,3) [L/S]	n/n/Y/n	+/+/+/-
Supertrend daily (10,3) [long-only]	Y/Y/Y/n	+/+/+/-
Time-series momentum 252d [L/S]	n/n/n/n	-/-/+/-
Time-series momentum 252d [long-only]	n/n/n/n	-/+/+/-
Trend(50) only [no OI]	n/n/n/n	+/+/+/-
Trend(N)+OI chg1 [expiry-adj OI]	n/n/n/n	-/+/+/-
Trend(N)+OI chg1 [naive OI]	n/n/n/Y	-/+/-/-
Trend(N)+OI chg5 [expiry-adj OI]	n/n/Y/n	+/+/+/-
Trend(N)+OI chg5 [naive OI]	n/n/n/n	+/+/+/-
Trend(N)+OI sma20 [expiry-adj OI]	n/n/n/n	+/+/+/-
Trend(N)+OI sma20 [naive OI]	n/n/n/Y	+/+/+/-

- **Positive in all four windows:** Bollinger mean-rev (20,2), IBS mean-reversion. (Bollinger mean-reversion has 0% exposure in later periods, so it is positive only because it barely trades.)
- **Beats buy-and-hold in Pre, Valid and Test together:** none.
- **Shortlist** (Sharpe above 0.5 and CAGR above 0 in train and valid, ranked by the weaker of the two): Supertrend daily (10,3) [long-only], Long build-up (price&OI up over k) [expiry-adj OI], Trend(N)+OI sma20 [expiry-adj OI], MACD 12/26/9 [long-only], EMA cross 12/26 [long-only], Keltner breakout (20,2).

4.3 Frozen picks (re-tuned, chosen before looking at test) and your strategy

Strategy	Pre 2013-17	Train 2013-21	Valid 2022-23	Test 2024-26	Full 2013-26	Exposure
Buy & Hold futures	4.5 / 0.33 / -27.4	7.4 / 0.48 / -40.7	6.0 / 0.48 / -17.8	-4.8 / -0.28 / -25.9	4.6 / 0.35 / -40.7	100%
Trend(50)+rising OI (5d), expiry-adjusted OI	-2.2 / -0.15 / -27.0	3.6 / 0.40 / -27.0	10.1 / 1.47 / -4.4	-8.5 / -1.15 / -23.9	1.9 / 0.25 / -27.0	45%
Trend(50)+rising OI (5d), naive OI	-1.4 / -0.11 / -17.6	1.1 / 0.17 / -17.6	5.5 / 0.93 / -3.6	-3.5 / -0.57 / -12.0	0.7 / 0.13 / -17.6	34%
Trend(50) only (no OI)	0.5 / 0.11 / -23.2	7.1 / 0.66 / -23.2	1.6 / 0.22 / -16.8	-4.6 / -0.50 / -15.7	3.8 / 0.40 / -23.2	59%
RE-TUNED LOCK: Supertrend daily (10,3) [long-only] {'w': 7, 'm': 2.0}	5.2 / 0.44 / -22.5	8.1 / 0.71 / -22.5	13.8 / 1.43 / -11.1	-6.1 / -0.71 / -19.1	5.8 / 0.57 / -22.5	55%
RE-TUNED LOCK: Long build-up (price&OI up over k) [expiry-adj OI] {'k': 20}	4.8 / 0.41 / -20.1	7.0 / 0.61 / -20.1	8.5 / 0.98 / -10.9	-9.4 / -1.12 / -25.7	3.6 / 0.37 / -25.7	55%
RE-TUNED LOCK: Trend(N)+OI sma20 [expiry-adj OI] {'N': 50}	0.4 / 0.09 / -23.2	6.3 / 0.60 / -23.2	5.8 / 0.72 / -10.9	-3.3 / -0.36 / -11.6	4.2 / 0.45 / -23.2	55%

All three re-tuned picks lose money in test (-3.3% to -9.4%), as the first report's picks did.

5. Trend(50) + rising OI

Strategy: long when close is above its 50-day average **and** total Nifty futures OI is higher than 5 trading days ago; flat otherwise.

5.1 Parameter sweep (N = 30, 40, 50, 60, 80) and three OI definitions, expiry-adjusted vs naive OI

Variant	N	Pre 2013-17	Train 2013-21	Valid 2022-23	Test 2024-26	Beats B&H (Pre/Valid/Test)
Trend(N)+OI chg5 [expiry-adj OI]	30	2.8 / 0.28 / -22.6	3.6 / 0.37 / -22.6	12.4 / 1.72 / -5.9	-10.3 / -1.35 / -28.8	n/Y/n
Trend(N)+OI chg5 [expiry-adj OI]	40	2.2 / 0.23 / -25.9	4.3 / 0.44 / -25.9	10.8 / 1.55 / -4.5	-9.5 / -1.22 / -26.6	n/Y/n
Trend(N)+OI chg5 [expiry-adj OI]	50	-2.2 / -0.15 / -27.0	3.6 / 0.40 / -27.0	10.1 / 1.47 / -4.4	-8.5 / -1.15 / -23.9	n/Y/n
Trend(N)+OI chg5 [expiry-adj OI]	60	-1.8 / -0.10 / -27.0	3.3 / 0.36 / -27.0	8.1 / 1.18 / -4.5	-7.0 / -0.92 / -19.3	n/Y/n
Trend(N)+OI chg5 [expiry-adj OI]	80	-0.5 / -0.01 / -20.8	3.2 / 0.38 / -20.8	11.0 / 1.55 / -5.1	-7.2 / -0.94 / -19.8	n/Y/n
Trend(N)+OI chg5 [naive OI]	30	-1.4 / -0.12 / -15.7	-0.5 / -0.02 / -15.7	6.8 / 1.11 / -5.3	-5.1 / -0.89 / -15.6	n/Y/n
Trend(N)+OI chg5 [naive OI]	40	0.1 / 0.06 / -16.2	1.2 / 0.18 / -16.2	5.9 / 0.99 / -4.6	-4.4 / -0.73 / -14.6	n/n/n
Trend(N)+OI chg5 [naive OI]	50	-1.4 / -0.11 / -17.6	1.1 / 0.17 / -17.6	5.5 / 0.93 / -3.6	-3.5 / -0.57 / -12.0	n/n/n
Trend(N)+OI chg5 [naive OI]	60	-0.6 / -0.02 / -15.9	0.8 / 0.13 / -15.9	3.8 / 0.64 / -4.0	-2.5 / -0.39 / -11.1	n/n/n
Trend(N)+OI chg5 [naive OI]	80	-1.0 / -0.06 / -15.3	-1.0 / -0.07 / -17.1	6.3 / 1.03 / -4.7	-3.2 / -0.49 / -11.5	n/Y/n
Trend(N)+OI chg1 [expiry-adj OI]	30	-2.2 / -0.16 / -22.6	-0.2 / 0.03 / -23.6	4.0 / 0.55 / -13.5	-7.2 / -1.05 / -20.0	n/n/n
Trend(N)+OI chg1 [expiry-adj OI]	40	-1.8 / -0.12 / -22.3	0.9 / 0.15 / -22.3	2.1 / 0.31 / -11.5	-4.9 / -0.69 / -14.4	n/n/n
Trend(N)+OI chg1 [expiry-adj OI]	50	-4.5 / -0.38 / -26.2	2.6 / 0.31 / -26.2	0.4 / 0.10 / -10.7	-2.9 / -0.42 / -12.3	n/n/n
Trend(N)+OI chg1 [expiry-adj OI]	60	-3.7 / -0.30 / -25.9	2.2 / 0.27 / -25.9	1.7 / 0.28 / -8.4	-1.4 / -0.18 / -8.2	n/n/Y
Trend(N)+OI chg1 [expiry-adj OI]	80	-3.5 / -0.35 / -24.5	1.4 / 0.20 / -24.5	3.6 / 0.53 / -7.8	-1.8 / -0.24 / -9.6	n/n/Y
Trend(N)+OI chg1 [naive OI]	30	0.2 / 0.07 / -12.8	0.8 / 0.13 / -24.4	0.7 / 0.13 / -13.5	-5.5 / -0.83 / -17.9	n/n/n
Trend(N)+OI chg1 [naive OI]	40	0.4 / 0.09 / -13.2	2.0 / 0.27 / -19.5	-0.3 / 0.00 / -12.0	-3.6 / -0.51 / -11.7	n/n/n
Trend(N)+OI chg1 [naive OI]	50	-2.4 / -0.22 / -18.9	3.0 / 0.38 / -18.9	-1.9 / -0.23 / -11.2	-0.8 / -0.10 / -10.9	n/n/Y
Trend(N)+OI chg1 [naive OI]	60	-1.6 / -0.13 / -18.7	2.6 / 0.33 / -18.7	-0.6 / -0.06 / -9.3	0.6 / 0.12 / -7.2	n/n/Y
Trend(N)+OI chg1 [naive OI]	80	-3.9 / -0.39 / -24.9	0.8 / 0.14 / -24.9	0.9 / 0.16 / -9.3	-0.2 / -0.00 / -8.7	n/n/Y
Trend(N)+OI sma20 [expiry-adj OI]	30	4.7 / 0.41 / -20.8	4.9 / 0.46 / -23.6	12.6 / 1.45 / -7.4	-5.1 / -0.60 / -16.5	Y/Y/n
Trend(N)+OI sma20 [expiry-adj OI]	40	4.7 / 0.41 / -23.1	6.4 / 0.57 / -23.1	8.5 / 0.99 / -9.7	-4.1 / -0.47 / -14.6	Y/Y/n
Trend(N)+OI sma20 [expiry-adj OI]	50	0.4 / 0.09 / -23.2	6.3 / 0.60 / -23.2	5.8 / 0.72 / -10.9	-3.3 / -0.36 / -11.6	n/n/n
Trend(N)+OI sma20 [expiry-adj OI]	60	-0.6 / 0.01 / -21.4	4.6 / 0.45 / -21.4	7.6 / 0.95 / -8.8	-1.9 / -0.20 / -10.2	n/Y/Y
Trend(N)+OI sma20 [expiry-adj OI]	80	0.3 / 0.08 / -19.1	4.4 / 0.46 / -19.1	9.7 / 1.19 / -7.4	-2.9 / -0.31 / -10.6	n/Y/n
Trend(N)+OI sma20 [naive OI]	30	0.1 / 0.05 / -15.0	0.3 / 0.08 / -15.0	5.8 / 0.90 / -4.7	-1.0 / -0.18 / -8.1	n/n/Y
Trend(N)+OI sma20 [naive OI]	40	1.3 / 0.20 / -13.3	2.1 / 0.30 / -13.3	3.9 / 0.61 / -6.9	-0.9 / -0.16 / -7.3	n/n/Y
Trend(N)+OI sma20 [naive OI]	50	0.4 / 0.09 / -12.7	2.2 / 0.31 / -12.7	1.3 / 0.23 / -7.8	-0.3 / -0.02 / -6.0	n/n/Y
Trend(N)+OI sma20 [naive OI]	60	0.1 / 0.06 / -13.0	0.5 / 0.10 / -13.0	0.6 / 0.13 / -8.7	-0.3 / -0.03 / -5.7	n/n/Y
Trend(N)+OI sma20 [naive OI]	80	0.4 / 0.09 / -10.0	-0.7 / -0.05 / -20.6	1.7 / 0.29 / -9.7	0.1 / 0.04 / -5.7	n/n/Y

Variant	N	Pre 2013-17	Train 2013-21	Valid 2022-23	Test 2024-26	Beats B&H (Pre/Valid/Test)
Trend(50) only [no OI]	30	5.2 / 0.44 / -20.8	5.9 / 0.53 / -20.8	7.3 / 0.81 / -16.2	-8.2 / -0.93 / -22.9	Y/Y/n
Trend(50) only [no OI]	40	5.3 / 0.45 / -23.1	7.5 / 0.65 / -23.1	4.5 / 0.51 / -15.0	-7.4 / -0.84 / -20.4	Y/n/n
Trend(50) only [no OI]	50	0.5 / 0.11 / -23.2	7.1 / 0.66 / -23.2	1.6 / 0.22 / -16.8	-4.6 / -0.50 / -15.7	n/n/n
Trend(50) only [no OI]	60	-0.4 / 0.03 / -22.3	5.9 / 0.55 / -22.3	2.4 / 0.31 / -14.9	-2.8 / -0.29 / -10.7	n/n/n
Trend(50) only [no OI]	80	0.4 / 0.10 / -20.8	5.0 / 0.49 / -20.8	5.9 / 0.69 / -7.7	-3.6 / -0.40 / -11.8	n/n/n

- **With the 5-day OI definition and expiry-adjusted OI, no N beats buy-and-hold in 2013–17 or in 2024–26.** The only strong window is 2022–23.
- **Adding OI to a plain trend filter did not help.** Trend(50)-only earned 0.5% in 2013–17 and the OI version -2.2%.
- **The few "Y" cells are not stable.** In 2013–17 they appear only for N=30/40 (sma20 OI, and Trend-only), and none of those hold up in the test window.

5.2 Year by year (net of costs)

Year	Buy & Hold	Trend(50)+rising OI (adj.)	Trend(50)+rising OI (naive)	Trend(50) only	Pair
2013	0.3%	-11.2%	-13.4%	-4.1%	2.9%
2014	22.1%	3.2%	8.2%	-0.9%	5.7%
2015	-10.4%	-2.2%	1.7%	-6.9%	2.8%
2016	-1.6%	-6.6%	-7.0%	5.1%	-5.6%
2017	23.5%	9.5%	2.2%	16.1%	3.6%
2018	0.1%	3.4%	9.9%	0.3%	-2.6%
2019	4.0%	-1.0%	2.1%	1.5%	-0.2%
2020	11.9%	27.4%	7.1%	45.1%	-0.6%
2021	21.1%	14.7%	1.3%	14.3%	9.6%
2022	0.5%	10.4%	4.4%	-7.5%	1.3%
2023	11.5%	9.3%	6.3%	11.5%	0.7%
2024	-0.4%	-14.5%	-2.6%	-4.2%	4.1%
2025	4.2%	-3.8%	-2.5%	1.9%	0.1%
2026	-16.0%	-5.2%	-4.8%	-10.1%	0.8%

The OI version has a large losing streak in 2013 (-11.2%), 2016 (-6.6%) and 2024 (-14.5%), all years in which buy-and-hold was flat (0.3%, -1.6%, -0.4%).

5.3 Expiry effect

OI signal	'OI rising' day after expiry	Other days	5d-change 'rising' in 5 days after expiry	Other days
Naive total OI	1%	65%	10%	73%
Expiry-adjusted OI	75%	65%	92%	73%

178 monthly expiries in 2012-2026; naive total OI falls 25% on average on the day after expiry.

- Total OI drops sharply after each monthly expiry because the expiring contract leaves the file, so a naive "OI rising" test is false almost every time on those days (1% vs 65%).

- An expiry-adjusted OI series (day-to-day change over contracts present on both days) removes the artifact. It slightly overstates "rising" right after expiry because the new far contract starts near zero (92% vs 73%).
- The handling choice changes the results (full-period CAGR 1.9% adjusted vs 0.7% naive), so both are shown everywhere.

5.4 Look-ahead check

Poison test: for 10 random cut dates every price/VIX/OI value after the cut was replaced by random numbers; 360 (strategy x cut) signals recomputed and compared up to the cut: 0 mismatches. Position at day h equals signal at close h-1: True.

OI for day t enters the signal at the close of t and the order is placed at the open of t+1, so the after-close publication of OI is respected.

6. Costs

Per side: brokerage 0.0012% + exchange 0.0019% + SEBI 0.0001% + 18% GST on those + slippage 0.0100%; stamp 0.0020% on buys; STT on sells 0.017% (to Sep 2013), 0.010% (Oct 2013-Mar 2023), 0.0125% (Apr 2023-Sep 2024), 0.020% (from Oct 2024). Round trip today: 0.050% of notional; roll = 0.060%.

Strategy	Gross full 2013-26	Net (slip 0.01%)	Net (slip 0.03%)	Round trips	Cost drag %/yr
Buy & Hold futures	5.3% / 0.38	4.6% / 0.35	4.1% / 0.32	0	0.67
Trend(50)+rising OI (5d), expiry-adjusted OI	2.8% / 0.34	1.9% / 0.25	1.0% / 0.16	212	0.93
Trend(50)+rising OI (5d), naive OI	1.7% / 0.26	0.7% / 0.13	-0.2% / 0.01	241	1.01
Trend(50) only (no OI)	4.5% / 0.47	3.8% / 0.40	3.2% / 0.35	113	0.71
RE-TUNED LOCK: Supertrend daily (10,3) [long-only] {'w': 7, 'm': 2.0}	6.4% / 0.62	5.8% / 0.57	5.3% / 0.52	92	0.64
RE-TUNED LOCK: Long build-up (price&OI up over k) [expiry-adj OI] {'k': 20}	4.5% / 0.45	3.6% / 0.37	2.8% / 0.30	168	0.89
RE-TUNED LOCK: Trend(N)+OI sma20 [expiry-adj OI] {'N': 50}	4.9% / 0.52	4.2% / 0.45	3.5% / 0.39	135	0.74

Trend(50)+rising OI makes 212 round trips over 13.8 years and costs roughly 0.9 points of CAGR per year. Costs are not what makes it lose; the signal does.

7. Overfitting accounting

Configurations evaluated in this run: 323; earlier work on 2018-26 data: 2354; N for deflation = 2677. Expected best annualised Sharpe from luck alone over N trials on the 2013-23 selection sample: 0.82.

Strategy	Train+valid 2013-23 Sharpe	Deflated Sharpe prob.	Test Sharpe	Test CAGR
Trend(50)+rising OI (5d), expiry-adjusted OI	0.53	0.17	-1.15	-8.5%
Trend(50)+rising OI (5d), naive OI	0.27	0.03	-0.57	-3.5%
Trend(50) only (no OI)	0.59	0.23	-0.50	-4.6%
RE-TUNED LOCK: Supertrend daily (10,3) [long-only] {'w': 7, 'm': 2.0}	0.82	0.49	-0.71	-6.1%
RE-TUNED LOCK: Long build-up (price&OI up over k) [expiry-adj OI] {'k': 20}	0.65	0.28	-1.12	-9.4%
RE-TUNED LOCK: Trend(N)+OI sma20 [expiry-adj OI] {'N': 50}	0.62	0.26	-0.36	-3.3%

All of these are far below the 0.95 that would be strong evidence. The trend/OI picks also lost money in test.

8. Walk-forward re-tuning (2015–2026)

Each year, pick the 5 best candidates by trailing Sharpe on data **before** that year, trade them equal-weight for the year.

Year	Picked (top 3 of 5 by trailing Sharpe)	WF return	Buy & Hold
2015	Connors RSI(2) 10/70 {'w': 3, 'lo': 5, 'hi'; IBS mean-reversion {'lo': 0.2, 'hi': 0.8}; IBS mean-reversion {'lo': 0.2, 'hi': 0.7}	4.5%	-10.4%
2016	Connors RSI(2) 10/70 {'w': 3, 'lo': 5, 'hi'; Connors RSI(2) 10/70 {'w': 3, 'lo': 5, 'hi'; IBS mean-reversion {'lo': 0.1, 'hi': 0.7}	-0.2%	-1.6%
2017	IBS mean-reversion {'lo': 0.1, 'hi': 0.7}; IBS mean-reversion {'lo': 0.2, 'hi': 0.8}; IBS mean-reversion {'lo': 0.2, 'hi': 0.7}	2.8%	23.5%
2018	IBS mean-reversion {'lo': 0.2, 'hi': 0.8}; IBS mean-reversion {'lo': 0.1, 'hi': 0.7}; IBS mean-reversion {'lo': 0.2, 'hi': 0.7}	-2.1%	0.1%
2019	Bollinger mean-rev (20,2) {'w': 20, 'k': 2.5}; IBS mean-reversion {'lo': 0.2, 'hi': 0.8}; IBS mean-reversion {'lo': 0.2, 'hi': 0.7}	7.0%	4.0%
2020	IBS mean-reversion {'lo': 0.2, 'hi': 0.8}; IBS mean-reversion {'lo': 0.2, 'hi': 0.7}; Stochastic mean-rev (14,20/80) {'w': 14, 'lo': 30, 'h	-8.8%	11.9%
2021	Bollinger breakout (20,2) {'w': 50, 'k': 2.0}; Bollinger breakout (20,2) {'w': 50, 'k': 2.5}; EMA cross 12/26 {'a': 8, 'b': 50}	12.0%	21.1%
2022	Bollinger breakout (20,2) {'w': 50, 'k': 2.0}; EMA cross 12/26 {'a': 20, 'b': 21}; EMA cross 12/26 {'a': 12, 'b': 26}	-1.3%	0.5%
2023	Bollinger breakout (20,2) {'w': 50, 'k': 2.0}; Supertrend daily (10,3) {'w': 7, 'm': 2.0}; EMA cross 12/26 {'a': 12, 'b': 26}	9.1%	11.5%
2024	Supertrend daily (10,3) {'w': 7, 'm': 2.0}; EMA cross 12/26 {'a': 12, 'b': 26}; EMA cross 12/26 {'a': 20, 'b': 21}	-6.5%	-0.4%
2025	EMA cross 12/26 {'a': 8, 'b': 50}; EMA cross 12/26 {'a': 12, 'b': 26}; Supertrend daily (10,3) {'w': 7, 'm': 2.0}	-2.3%	4.2%
2026	Supertrend daily (10,3) {'w': 7, 'm': 2.0}; EMA cross 12/26 {'a': 8, 'b': 50}; EMA cross 12/26 {'a': 12, 'b': 26}	-5.3%	-16.0%

Series (2015-2026, out-of-sample years only)	CAGR	Sharpe	MaxDD
Walk-forward top-5 (re-tuned yearly)	0.6%	0.11	-29.4%
Buy & Hold futures	3.6%	0.29	-40.7%
Panic-reversal pair (post-hoc, fixed rules)	1.1%	0.36	-12.9%

Automatic yearly re-selection from this menu earns less than buy-and-hold and has a -29% drawdown. The picks flip between mean-reversion (2015–2020) and trend (2021–2026), which is exactly when each fails.

9. Regime-wise results (annualised return, Sharpe in brackets; `*` = fewer than 30 days)

Regime	Period	Days	Buy & Hold	Trend(50)+rising OI (adj.)	Trend(50) only	Supertrend daily (10,3) LO	Long build-up (price&OI up ove	Trend(N)+OI sma20 [expiry-adj
Trend (price vs 200DMA): Bear	pre	446	7.3% (Sh 0.4)	-2.0% (Sh -0.3)	5.4% (Sh 0.7)	12.5% (Sh 1.3)	8.8% (Sh 0.9)	7.1% (Sh 0.9)
Trend (price vs 200DMA): Bear	valid	163	24.1% (Sh 1.4)	0.1% (Sh 0.0)	7.9% (Sh 1.3)	23.9% (Sh 2.7)	9.1% (Sh 1.3)	4.0% (Sh 0.8)
Trend (price vs 200DMA): Bear	test	340	-0.8% (Sh -0.0)	-0.7% (Sh -0.2)	2.0% (Sh 0.3)	4.1% (Sh 0.5)	-6.4% (Sh -0.9)	6.0% (Sh 1.1)
Trend (price vs 200DMA): Bull	pre	599	5.2% (Sh 0.3)	-1.4% (Sh -0.1)	-1.8% (Sh -0.1)	1.1% (Sh 0.1)	3.2% (Sh 0.2)	-3.4% (Sh -0.2)
Trend (price vs 200DMA): Bull	valid	329	-1.6% (Sh -0.1)	14.7% (Sh 1.9)	-0.9% (Sh -0.1)	8.1% (Sh 0.9)	8.2% (Sh 0.9)	7.0% (Sh 0.7)
Trend (price vs 200DMA): Bull	test	363	-7.0% (Sh -0.6)	-16.1% (Sh -1.7)	-10.2% (Sh -1.0)	-15.4% (Sh -1.7)	-12.3% (Sh -1.3)	-11.4% (Sh -1.1)
Volatility (India VIX): High >22	pre	99	43.7% (Sh 1.7)	22.2% (Sh 1.6)	28.1% (Sh 1.7)	18.5% (Sh 1.1)	31.9% (Sh 2.0)	28.1% (Sh 1.7)
Volatility (India VIX): High >22	valid	48	16.1% (Sh 0.6)	-6.0% (Sh -2.3)	-7.4% (Sh -1.3)	27.1% (Sh 2.8)	1.5% (Sh 0.3)	-7.4% (Sh -1.3)

Regime	Period	Days	Buy & Hold	Trend(50)+rising OI (adj.)	Trend(50) only	Supertrend daily (10,3) LO	Long build-up (price&OI up ove	Trend(N)+OI sma20 [expiry-adj
Volatility (India VIX): High >22	test	20	98%*	-56%*	-56%*	-29%*	-29%*	-56%*
Volatility (India VIX): Low <15	pre	297	-15.8% (Sh -1.0)	-9.2% (Sh -0.7)	-10.2% (Sh -0.8)	-10.7% (Sh -0.8)	-13.1% (Sh -1.0)	-12.9% (Sh -1.0)
Volatility (India VIX): Low <15	valid	249	7.2% (Sh 0.7)	9.4% (Sh 1.5)	10.2% (Sh 1.4)	17.7% (Sh 2.6)	10.7% (Sh 1.5)	12.9% (Sh 1.8)
Volatility (India VIX): Low <15	test	491	-20.0% (Sh -1.6)	-7.2% (Sh -1.1)	-7.2% (Sh -1.0)	-8.5% (Sh -1.1)	-13.6% (Sh -1.7)	-5.3% (Sh -0.7)
Volatility (India VIX): Mid 15-22	pre	649	10.4% (Sh 0.6)	-1.8% (Sh -0.2)	2.5% (Sh 0.2)	11.7% (Sh 0.9)	10.1% (Sh 0.8)	3.4% (Sh 0.3)
Volatility (India VIX): Mid 15-22	valid	195	4.2% (Sh 0.3)	14.3% (Sh 1.8)	-6.2% (Sh -0.5)	4.3% (Sh 0.4)	7.4% (Sh 0.7)	0.5% (Sh 0.0)
Volatility (India VIX): Mid 15-22	test	192	26.4% (Sh 1.7)	-7.5% (Sh -0.8)	8.5% (Sh 0.8)	2.8% (Sh 0.3)	3.2% (Sh 0.3)	8.5% (Sh 0.8)
Momentum (20d return): Down <-3%	pre	231	12.8% (Sh 0.6)	0.0% (Sh 0.0)	-0.1% (Sh -1.8)	7.9% (Sh 1.2)	0.0% (Sh 0.0)	-0.1% (Sh -1.5)
Momentum (20d return): Down <-3%	valid	102	16.2% (Sh 0.9)	-0.1% (Sh -1.6)	-4.7% (Sh -1.6)	1.3% (Sh 0.2)	0.0% (Sh 0.0)	-4.7% (Sh -1.6)
Momentum (20d return): Down <-3%	test	137	-8.5% (Sh -0.5)	0.0% (Sh 0.0)	0.0% (Sh 0.0)	1.1% (Sh 0.4)	0.0% (Sh 0.0)	0.0% (Sh 0.0)
Momentum (20d return): Flat	pre	511	2.3% (Sh 0.1)	-4.3% (Sh -0.3)	-1.2% (Sh -0.1)	5.3% (Sh 0.3)	6.6% (Sh 0.4)	-2.3% (Sh -0.2)
Momentum (20d return): Flat	valid	249	-5.8% (Sh -0.5)	-0.8% (Sh -0.1)	-8.0% (Sh -1.0)	9.3% (Sh 1.1)	0.5% (Sh 0.1)	-3.6% (Sh -0.5)
Momentum (20d return): Flat	test	467	4.4% (Sh 0.4)	-3.8% (Sh -0.5)	-0.8% (Sh -0.1)	-2.6% (Sh -0.3)	-7.6% (Sh -0.9)	0.4% (Sh 0.1)
Momentum (20d return): Up >3%	pre	303	7.3% (Sh 0.5)	1.7% (Sh 0.1)	6.6% (Sh 0.5)	5.7% (Sh 0.4)	8.1% (Sh 0.6)	7.7% (Sh 0.6)
Momentum (20d return): Up >3%	valid	141	22.6% (Sh 1.6)	35.8% (Sh 3.8)	24.6% (Sh 1.9)	29.1% (Sh 2.3)	28.8% (Sh 2.2)	30.8% (Sh 2.6)
Momentum (20d return): Up >3%	test	99	-37.1% (Sh -2.3)	-43.3% (Sh -3.7)	-26.8% (Sh -2.0)	-32.0% (Sh -2.3)	-31.5% (Sh -2.3)	-23.2% (Sh -1.7)
Drawdown from high: DD >10%	pre	435	18.3% (Sh 1.0)	0.8% (Sh 0.1)	8.2% (Sh 0.9)	15.8% (Sh 1.6)	8.7% (Sh 0.9)	7.6% (Sh 0.9)
Drawdown from high: DD >10%	valid	104	35.4% (Sh 1.9)	-1.7% (Sh -1.8)	8.5% (Sh 1.6)	18.3% (Sh 1.8)	1.2% (Sh 0.2)	1.3% (Sh 0.4)
Drawdown from high: DD >10%	test	384	0.1% (Sh 0.0)	0.3% (Sh 0.1)	2.3% (Sh 0.4)	4.7% (Sh 0.6)	-3.7% (Sh -0.5)	6.1% (Sh 1.1)
Drawdown from high: Near high	pre	610	-2.6% (Sh -0.1)	-3.3% (Sh -0.3)	-3.7% (Sh -0.3)	-1.0% (Sh -0.1)	3.3% (Sh 0.2)	-3.6% (Sh -0.3)
Drawdown from high: Near high	valid	388	-0.7% (Sh -0.1)	12.9% (Sh 1.7)	0.3% (Sh 0.0)	12.0% (Sh 1.3)	10.5% (Sh 1.2)	7.3% (Sh 0.8)
Drawdown from high: Near high	test	319	-8.8% (Sh -0.7)	-19.4% (Sh -1.9)	-12.2% (Sh -1.1)	-18.8% (Sh -2.0)	-16.4% (Sh -1.6)	-14.0% (Sh -1.3)

What the regimes show:

- **Trend strategies gain in "up >3%" and bull regimes in some windows and lose in the next.** The sign flips between windows, so these are not stable edges.
- **The OI strategy does not protect in high-VIX or drawdown regimes** in the windows that have enough days.
- **Supertrend (7, 2.0) long-only is positive in bear-trend regimes in all three windows** (12.5%, 23.9%, 4.1%) and loses in the test-period bull regime (-15.4%). It is a description of what happened, not a tested rule.
- High-VIX days are rare in test (20 days), so single numbers there are noisy.

10. The one family that stayed positive: IBS mean-reversion

Rule: long when the day's close is in the bottom 10% of the day's high-low range (IBS below 0.1); exit when IBS rises above 0.8. Parameters (0.1 / 0.8) come from the earlier 2018–21 tuning and were **not** changed after seeing 2013–17 or test.

Strategy	Pre 2013-17	Train 2013-21	Valid 2022-23	Test 2024-26	Full 2013-26	Exposure
IBS mean-reversion {'lo': 0.1, 'hi': 0.8} (frozen from 2018-21)	6.0 / 0.82 / -5.6	3.6 / 0.47 / -13.8	1.1 / 0.25 / -6.1	3.1 / 0.50 / -7.9	3.2 / 0.45 / -13.8	17%

Strategy	Pre 2013-17	Train 2013-21	Valid 2022-23	Test 2024-26	Full 2013-26	Exposure
Buy & Hold futures	4.5 / 0.33 / -27.4	7.4 / 0.48 / -40.7	6.0 / 0.48 / -17.8	-4.8 / -0.28 / -25.9	4.6 / 0.35 / -40.7	100%
Panic-reversal pair (post-hoc)	1.3 / 0.50 / -6.5	1.7 / 0.48 / -12.9	1.0 / 0.50 / -3.1	1.8 / 0.66 / -2.0	1.6 / 0.50 / -12.9	7%

10.1 Robustness to its two parameters (CAGR% in Pre / Valid / Test)

lo \ hi	hi=0.6	hi=0.7	hi=0.8	hi=0.9
lo=0.05	Pre 0.5 / Val 0.4 / Te 0.6	Pre 0.3 / Val 0.4 / Te 0.6	Pre 0.7 / Val 1.3 / Te 0.6	Pre 0.8 / Val 1.3 / Te 0.6
lo=0.1	Pre 6.4 / Val 0.5 / Te 3.5	Pre 7.5 / Val 0.1 / Te 1.1	Pre 6.0 / Val 1.1 / Te 3.1	Pre 6.8 / Val 8.1 / Te -3.8
lo=0.15	Pre 5.4 / Val 2.0 / Te 5.0	Pre 7.1 / Val 1.3 / Te 3.9	Pre 8.2 / Val 0.4 / Te 1.9	Pre 6.6 / Val 3.1 / Te -6.2
lo=0.2	Pre 10.8 / Val -1.7 / Te 2.3	Pre 12.1 / Val -1.4 / Te 0.7	Pre 13.7 / Val -3.1 / Te -0.9	Pre 6.7 / Val -0.2 / Te -8.6
lo=0.25	Pre 7.2 / Val 0.7 / Te 1.6	Pre 8.3 / Val -1.4 / Te -0.4	Pre 10.5 / Val -2.8 / Te -1.4	Pre 5.3 / Val 1.0 / Te -6.4
lo=0.3	Pre 6.2 / Val 3.4 / Te -0.6	Pre 6.9 / Val 1.1 / Te -2.8	Pre 9.5 / Val 0.0 / Te -3.5	Pre 3.7 / Val 3.5 / Te -7.3

- Around lo=0.10 to 0.15 the strategy is positive in most windows. At lo=0.20 and above it fails in test, and at lo=0.05 it hardly trades. This is a modest zone of stability, not a plateau.

10.2 Year by year, and combined with the panic-reversal pair

Year	Buy & Hold	IBS mean-reversion	Pair	IBS + Pair
2013	0.3%	4.9%	2.9%	3.9%
2014	22.1%	5.3%	5.7%	5.5%
2015	-10.4%	10.7%	2.8%	6.7%
2016	-1.6%	3.2%	-5.6%	-1.2%
2017	23.5%	0.1%	3.6%	1.8%
2018	0.1%	-4.4%	-2.6%	-3.4%
2019	4.0%	2.9%	-0.2%	1.4%
2020	11.9%	-0.5%	-0.6%	-0.4%
2021	21.1%	10.9%	9.6%	10.3%
2022	0.5%	3.3%	1.3%	2.3%
2023	11.5%	-1.0%	0.7%	-0.1%
2024	-0.4%	3.4%	4.1%	3.8%
2025	4.2%	7.3%	0.1%	3.7%
2026	-16.0%	-1.8%	0.8%	-0.4%

Series	Pre 2013-17	Train 2013-21	Valid 2022-23	Test 2024-26	Full 2013-26
IBS + Pair, 50/50	3.7 / 0.87 / -4.2	2.7 / 0.54 / -12.1	1.1 / 0.41 / -3.1	2.5 / 0.70 / -3.8	2.4 / 0.54 / -12.1
IBS alone	6.0 / 0.82 / -5.6	3.6 / 0.47 / -13.8	1.1 / 0.25 / -6.1	3.1 / 0.50 / -7.9	3.2 / 0.45 / -13.8
Buy & Hold	4.5 / 0.33 / -27.4	7.4 / 0.48 / -40.7	6.0 / 0.48 / -17.8	-4.8 / -0.28 / -25.9	4.6 / 0.35 / -40.7

- IBS is positive in 10 of 14 years; it loses in 2018, 2020 (-0.5%), 2023 and 2026 and trails buy-and-hold in strong bull years.
- A 50/50 mix with the panic-reversal pair (VIX spike or pullback in an uptrend) smooths the path: full-period 2.4% CAGR, Sharpe 0.54, MaxDD -12.1%, and positive in all five windows.

10.3 Costs and significance

Slippage per side	Full-period CAGR / Sharpe / MaxDD	Round trips
0.00%	3.5% / 0.48 / -13.1%	154
0.01%	3.2% / 0.45 / -13.8%	154
0.03%	2.6% / 0.37 / -15.2%	154
0.05%	2.0% / 0.30 / -16.6%	154

IBS: train+valid(2013-23) Sharpe 0.44, Deflated Sharpe probability 0.10 (N=2677).

10.4 Regime-wise

Regime	Period	Days	Buy & Hold	IBS mean-reversion	Pair
Trend (price vs 200DMA): Bear	pre	446	7.3% (Sh 0.4)	11.1% (Sh 1.3)	-0.8% (Sh -0.4)
Trend (price vs 200DMA): Bear	train_old	266	9.5% (Sh 0.3)	0.4% (Sh 0.0)	-1.2% (Sh -0.2)
Trend (price vs 200DMA): Bear	valid	163	24.1% (Sh 1.4)	10.2% (Sh 1.8)	0.6% (Sh 0.2)
Trend (price vs 200DMA): Bear	test	340	-0.8% (Sh -0.0)	5.4% (Sh 0.7)	0.6% (Sh 0.7)
Trend (price vs 200DMA): Bull	pre	599	5.2% (Sh 0.3)	2.3% (Sh 0.4)	2.9% (Sh 0.9)
Trend (price vs 200DMA): Bull	train_old	661	13.7% (Sh 1.0)	4.0% (Sh 0.6)	3.5% (Sh 0.9)
Trend (price vs 200DMA): Bull	valid	329	-1.6% (Sh -0.1)	-3.2% (Sh -0.7)	1.3% (Sh 0.8)
Trend (price vs 200DMA): Bull	test	363	-7.0% (Sh -0.6)	1.3% (Sh 0.3)	2.9% (Sh 0.8)
Volatility (India VIX): High >22	pre	99	43.7% (Sh 1.7)	27.2% (Sh 2.6)	4.5% (Sh 0.9)
Volatility (India VIX): High >22	train_old	205	31.4% (Sh 0.9)	10.9% (Sh 0.6)	10.5% (Sh 1.2)
Volatility (India VIX): High >22	valid	48	16.1% (Sh 0.6)	12.6% (Sh 4.9)	0.0% (Sh 0.0)
Volatility (India VIX): High >22	test	20	98%*	21%*	48%*
Volatility (India VIX): Low <15	pre	297	-15.8% (Sh -1.0)	-4.7% (Sh -0.8)	1.5% (Sh 0.9)
Volatility (India VIX): Low <15	train_old	278	2.0% (Sh 0.2)	-0.9% (Sh -0.2)	-1.7% (Sh -1.2)
Volatility (India VIX): Low <15	valid	249	7.2% (Sh 0.7)	-1.9% (Sh -0.5)	0.4% (Sh 0.5)
Volatility (India VIX): Low <15	test	491	-20.0% (Sh -1.6)	-3.3% (Sh -0.6)	-0.0% (Sh -0.0)
Volatility (India VIX): Mid 15-22	pre	649	10.4% (Sh 0.6)	7.8% (Sh 1.1)	0.8% (Sh 0.3)
Volatility (India VIX): Mid 15-22	train_old	444	10.3% (Sh 0.6)	1.8% (Sh 0.2)	0.7% (Sh 0.2)
Volatility (India VIX): Mid 15-22	valid	195	4.2% (Sh 0.3)	2.5% (Sh 0.4)	2.1% (Sh 1.1)
Volatility (India VIX): Mid 15-22	test	192	26.4% (Sh 1.7)	18.4% (Sh 2.1)	1.7% (Sh 0.9)
Momentum (20d return): Down <-3%	pre	231	12.8% (Sh 0.6)	22.9% (Sh 2.1)	1.8% (Sh 0.4)
Momentum (20d return): Down <-3%	train_old	142	-22.3% (Sh -0.6)	11.2% (Sh 0.6)	8.4% (Sh 0.8)

Regime	Period	Days	Buy & Hold	IBS mean-reversion	Pair
Momentum (20d return): Down <-3%	valid	102	16.2% (Sh 0.9)	13.6% (Sh 2.0)	7.5% (Sh 2.8)
Momentum (20d return): Down <-3%	test	137	-8.5% (Sh -0.5)	6.5% (Sh 0.6)	-0.6% (Sh -0.3)
Momentum (20d return): Flat	pre	511	2.3% (Sh 0.1)	2.2% (Sh 0.3)	1.6% (Sh 0.8)
Momentum (20d return): Flat	train_old	449	15.8% (Sh 1.1)	1.2% (Sh 0.2)	1.2% (Sh 0.4)
Momentum (20d return): Flat	valid	249	-5.8% (Sh -0.5)	-2.2% (Sh -0.7)	-1.0% (Sh -0.4)
Momentum (20d return): Flat	test	467	4.4% (Sh 0.4)	3.7% (Sh 0.7)	2.9% (Sh 0.9)
Momentum (20d return): Up >3%	pre	303	7.3% (Sh 0.5)	-0.1% (Sh -0.0)	0.5% (Sh 0.7)
Momentum (20d return): Up >3%	train_old	336	22.7% (Sh 1.5)	2.0% (Sh 0.3)	0.7% (Sh 0.5)
Momentum (20d return): Up >3%	valid	141	22.6% (Sh 1.6)	-1.5% (Sh -0.2)	0.0% (Sh 0.0)
Momentum (20d return): Up >3%	test	99	-37.1% (Sh -2.3)	-3.2% (Sh -0.8)	0.0% (Sh 0.0)
Drawdown from high: DD >10%	pre	435	18.3% (Sh 1.0)	11.9% (Sh 1.4)	-2.9% (Sh -1.0)
Drawdown from high: DD >10%	train_old	224	20.5% (Sh 0.6)	9.6% (Sh 0.6)	3.7% (Sh 0.5)
Drawdown from high: DD >10%	valid	104	35.4% (Sh 1.9)	12.1% (Sh 1.8)	4.4% (Sh 1.9)
Drawdown from high: DD >10%	test	384	0.1% (Sh 0.0)	6.2% (Sh 0.8)	0.8% (Sh 0.8)
Drawdown from high: Near high	pre	610	-2.6% (Sh -0.1)	2.0% (Sh 0.3)	4.3% (Sh 1.8)
Drawdown from high: Near high	train_old	703	9.9% (Sh 0.7)	0.9% (Sh 0.1)	1.6% (Sh 0.5)
Drawdown from high: Near high	valid	388	-0.7% (Sh -0.1)	-1.6% (Sh -0.4)	0.1% (Sh 0.1)
Drawdown from high: Near high	test	319	-8.8% (Sh -0.7)	-0.2% (Sh -0.1)	3.0% (Sh 0.8)

- **IBS earns in stressed markets:** high VIX (27.2%, 10.9%, 12.6%), drawdown above 10% (11.9%, 9.6%, 12.1%, 6.2%) and 20-day drop below -3% (22.9%, 11.2%, 13.6%, 6.5%) are positive in every window that has enough days.
- **It loses in calm, rising markets:** low VIX below 15 is negative in all four windows (-4.7%, -0.9%, -1.9%, -3.3%), and "up >3%" is about zero or negative.

10.5 Regime-conditioned IBS: hold only when VIX is at least 15

The 15 boundary is the Low/Mid VIX edge I had fixed in advance. **Caveat:** I chose to test a VIX filter after looking at the regime table above, which includes valid and test windows, so this is partly in-sample. Treat it as a hypothesis; the old-train column (2018–21) is the only fully unseen check on the rule, and the sweep shows how sensitive the threshold is.

VIX filter	Pre 2013-17	Old train 2018-21	Valid 2022-23	Test 2024-26	Full 2013-26	Exposure
none (plain IBS)	6.0 / 0.82 / -5.6	2.6 / 0.31 / -11.7	1.1 / 0.25 / -6.1	3.1 / 0.50 / -7.9	3.2 / 0.45 / -13.8	17%
VIX >= 12	6.0 / 0.82 / -5.6	2.6 / 0.31 / -11.7	1.6 / 0.37 / -6.1	3.4 / 0.56 / -7.9	3.3 / 0.47 / -13.6	16%
VIX >= 14	7.5 / 1.07 / -5.0	3.2 / 0.37 / -11.6	1.3 / 0.31 / -6.1	4.6 / 0.85 / -5.2	4.2 / 0.61 / -11.6	13%
VIX >= 15	7.4 / 1.10 / -5.0	2.8 / 0.34 / -11.6	2.1 / 0.49 / -6.1	5.5 / 1.12 / -3.6	4.3 / 0.65 / -12.2	11%
VIX >= 16	6.3 / 0.97 / -5.0	2.6 / 0.32 / -12.8	0.6 / 0.17 / -6.1	1.3 / 0.36 / -3.6	3.0 / 0.48 / -12.8	9%
VIX >= 18	7.0 / 1.27 / -3.2	1.5 / 0.21 / -10.9	2.1 / 0.60 / -3.2	0.5 / 0.22 / -2.1	2.8 / 0.50 / -13.0	5%
VIX >= 20	7.0 / 1.44 / -2.0	3.0 / 0.40 / -9.4	2.9 / 1.20 / -1.3	0.0 / 0.03 / -2.1	3.3 / 0.65 / -9.4	3%

- Thresholds from 12 to 15 improve on plain IBS in every window. From 16 upward the strategy trades rarely (9% of days or less) and the test window fades, so the benefit is concentrated near 14–15.

- Exposure falls from 17% to 11% of days, with 118 round trips in 13.8 years (about 9 a year).

Year	Buy & Hold	IBS plain	IBS (VIX>=15)
2013	0.3%	4.9%	5.6%
2014	22.1%	5.3%	7.0%
2015	-10.4%	10.7%	14.4%
2016	-1.6%	3.2%	4.1%
2017	23.5%	0.1%	1.0%
2018	0.1%	-4.4%	-5.3%
2019	4.0%	2.9%	4.6%
2020	11.9%	-0.5%	-0.9%
2021	21.1%	10.9%	10.1%
2022	0.5%	3.3%	3.0%
2023	11.5%	-1.0%	1.0%
2024	-0.4%	3.4%	7.3%
2025	4.2%	7.3%	1.7%
2026	-16.0%	-1.8%	6.5%

Series	Pre 2013-17	Train 2013-21	Valid 2022-23	Test 2024-26	Full 2013-26
IBS (VIX>=15) + Pair, 50/50	4.4 / 1.12 / -4.1	3.1 / 0.64 / -11.5	1.6 / 0.66 / -3.1	3.7 / 1.29 / -1.7	3.0 / 0.71 / -11.5
IBS (VIX>=15) alone	7.4 / 1.10 / -5.0	4.5 / 0.59 / -12.2	2.1 / 0.49 / -6.1	5.5 / 1.12 / -3.6	4.3 / 0.65 / -12.2
Buy & Hold	4.5 / 0.33 / -27.4	7.4 / 0.48 / -40.7	6.0 / 0.48 / -17.8	-4.8 / -0.28 / -25.9	4.6 / 0.35 / -40.7

Slippage per side	Full CAGR / Sharpe / MaxDD
0.00%	4.5% / 0.67 / -11.6%
0.01%	4.3% / 0.65 / -12.2%
0.03%	3.9% / 0.59 / -13.2%
0.05%	3.5% / 0.53 / -14.2%

IBS (VIX>=15): train+valid Sharpe 0.57, Deflated Sharpe probability 0.19 (N=2684). The probability is still far below 0.95: the filtered version is not proven either.

11. Post-hoc panic-reversal pair (kept for completeness)

Found earlier with knowledge of 2018–2026, so **2013–17 is genuinely out-of-sample for it**: it returned 1.3% (Sharpe 0.50, MaxDD -6.5%) there, similar to its 2018–26 results.

Strategy	Pre 2013-17	Train 2013-21	Valid 2022-23	Test 2024-26	Full 2013-26	Exposure
VIX spike >12% (long 1 day)	-0.1 / -0.03 / -5.7	0.4 / 0.10 / -14.6	-0.5 / -0.13 / -4.6	1.1 / 0.37 / -2.3	0.4 / 0.11 / -14.6	2%
10-day drop >2.5% in uptrend (long 1 day)	2.7 / 0.64 / -8.1	2.9 / 0.60 / -12.2	2.5 / 1.06 / -1.9	2.5 / 0.79 / -4.0	2.7 / 0.65 / -12.2	5%
Pair (equal weight)	1.3 / 0.50 / -6.5	1.7 / 0.48 / -12.9	1.0 / 0.50 / -3.1	1.8 / 0.66 / -2.0	1.6 / 0.50 / -12.9	7%

12. Can this work in real life? (honest answer)

- **Trend(50) + rising OI: no.** It failed on two independent windows after costs, and fine-tuning did not fix it.
- **IBS mean-reversion (alone, filtered by VIX, or with the pair): maybe, but small.** Expect roughly 3–5% a year on notional, with a worst historical drawdown around -12% to -14%, and long stretches behind buy-and-hold in bull markets. The VIX-filtered version had 4.3% CAGR / 0.65 Sharpe / -12.2% MaxDD over 2013–26 and 3.0% / 0.71 / -11.5% when mixed 50/50 with the pair. None of this is statistically proven after the number of trials tried (Deflated Sharpe 0.10 to 0.19).
- **Gate you set** ("beat buy-and-hold after costs"): IBS beats it on drawdown and Sharpe in the full period and wins in 2013–17 and 2024–26, but earns less than buy-and-hold in 2018–23. Whether that passes depends on whether you judge by total return or by risk-adjusted return.
- **Paper trading first** is the right next step for IBS. It trades about 11 times a year, so 2–3 months gives only a handful of trades. Use it to check fills, gaps at the open and signal timing, not to judge profits.
- **Leverage is dangerous.** The strategies are mostly flat, so their return per rupee of margin looks higher, but drawdowns scale with leverage and March 2020 is in the data.

13. Limitations

- Daily data only; fills at the open plus 0.01% slippage per side. Real fills at volatile opens can be worse.
- One instrument. 14 years includes the 2020 crash only once and few trending bull years after 2021.
- No margin, lot-size, funding or tax modeling. Returns are on notional exposure.
- Total OI includes calendar-spread and hedging positions, so it is not a clean directional-positioning measure.
- **Test period (2024–26) is no longer completely untouched**: earlier exploratory runs looked at it. The 2013–17 window is the clean out-of-sample check for the frozen picks.
- Regime thresholds (VIX 15/22, momentum +/-3%, drawdown 10%) were fixed once and not tuned; many small regime cells are noisy.

14. Files

- Code: `scripts/report_futures_ext.py` (all results), `scripts/build_report_ext.py` (this document); earlier `scripts/report_futures.py`
- Data: `data/fut/nifty_fut_2012_2017.csv` (new), `data/vix_ext.csv`; raw copy `data/fut/nifty_fut_2012_2017_raw.csv`
- Results: `results/report_ext_tables.json`, `results/report_ext_res.pkl`